

Sarah Binta Alam Shoilee

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Research Experience

- 02/2026 – 07/2026 📌 **Researcher-in-Residence, Koninklijke Bibliotheek (KB), the Netherlands.**
- This project aims to explore the use of Named Entity Recognition and Entity Linking to connect historical Dutch newspaper data from the Delpher corpus with museum collector records from the Nationaal Museum van Wereldculturen and Museum Bronbeek. It focuses on developing and evaluating linking strategies for incomplete, ambiguous, and biased datasets, with particular attention to long-tail and under-documented individuals.
- 09/2021 – 04/2026 📌 **Ph.D. Researcher, Vrije Universiteit Amsterdam, the Netherlands.**
- My PhD project developed provenance-aware Semantic Web and Knowledge Graph methods to support knowledge discovery in colonial heritage collections. The project focused on semantic data modelling, entity alignment, exploratory visual analytics, and multimodal knowledge graph enrichment from archival sources, with an emphasis on transparency, trust, and human-centred AI for provenance research.
Promoters: Prof. Dr. Jacco van Ossenbruggen & Prof. Dr. Susan Legêne
Co-promoter: Dr. Victor de Boer
- 02/2020 – 06/2020 📌 **UX Research Trainee, Hogeschool van Amsterdam, the Netherlands.**
- Designed and evaluated user-interface for an explainable AI-driven web platform for promoting healthy lifestyles, through user-centered studies in collaboration with TNO.
- 06/2017 – 08/2018 📌 **Research Assistant, North South University, Bangladesh.**
- Conducted focus groups and qualitative analysis for *Google's Next Billion Users Initiative*, proposing solutions to enhance product adoption among tech users in Bangladesh.
- Designed and evaluated a mobile application interface for breast cancer awareness, focusing on user satisfaction and sustainable adoption.

Teaching Experience

- 2022 – 2024 📌 **Thesis Supervisor, Vrije Universiteit Amsterdam, the Netherlands.**
- Independently supervised one Bachelor thesis in Computer Science.
- Co-supervised two Master's theses in Information Studies, one in Artificial Intelligence and one Bachelor project in Artificial Intelligence.
- 2021 – 2023 📌 **Teaching Assistant, Vrije Universiteit Amsterdam, the Netherlands.**
Grading assistance for the following course:
- Knowledge Organization
- 2017 – 2018 📌 **Teaching Assistant, North South University, Bangladesh.**
- Instructed and conducted practical sessions for undergraduate students in courses: Database Management, Digital Logic Design and Web Development

Education

- 09/2021 – 04/2026  **Ph.D. in Computer Science, Vrije Universiteit Amsterdam**, the Netherlands.
Thesis Title: *Making the Invisible Visible: A Semantic Web Approach for Knowledge Discovery in Colonial Collections*
- 09/2018 – 08/2020  **M.Sc. in Artificial Intelligence, KU Leuven**, Belgium.
Specialization: Big Data Analytics
- 09/2016 – 12/2017  **M.Sc. in Computer Science & Engineering, North South University**, Bangladesh.  *Summa Cum Laude*
- 2012/01 – 2016/08  **B.Sc. in Computer Science & Engineering, North South University**, Bangladesh.

Research Publications

Conference Proceedings

- 1 S. B. A. Shoilee, V. de Boer, J. van Ossenbruggen, and S. Legêne, “From historical tabular image to knowledge graphs: A provenance-aware modular pipeline,” in *Proceedings of the 5th International Conference on Hybrid Human-Artificial Intelligence (HHAI 2026)*, Forthcoming, IOS Press, 2026.
- 2 S. B. A. Shoilee et al., “Bridging data gaps: Harnessing semantic associations for knowledge discovery in colonial heritage,” in *Proceedings of the Knowledge Capture Conference 2025 (K-CAP ’25)*, Dayton, OH, USA: ACM, 2025, pp. 1–9.
- 3 S. B. A. Shoilee et al., “Enhancing provenance research with linked data: A visual approach to knowledge discovery,” in *Proceedings: SemDH 2025 Second International Workshop of Semantic Digital Humanities, co-located with ESWC 2025, Portoroz, Slovenia*,  Best Paper, CEUR Workshop Proceedings, 2025.
- 4 S. B. A. Shoilee et al., “PM-Sampo: Semantic portal for heritage object provenance research,” in *The Semantic Web: ESWC 2025 Satellite Events, Portoroz, Slovenia, June 1 - 5, 2025, Proceedings*,  Nominated for Best Demo, Springer-Verlag, 2025.
- 5 S. B. A. Shoilee, V. de Boer, and J. van Ossenbruggen, “A framework for evaluating entity alignment impact on downstream knowledge discovery,” in *International Conference on Knowledge Engineering and Knowledge Management*, Springer, 2024, pp. 226–242.
- 6 S. B. A. Shoilee, V. de Boer, and J. van Ossenbruggen, “Polyvocal knowledge modelling for ethnographic heritage object provenance,” in *Knowledge graphs: Semantics, machine learning, and languages*, IOS Press, 2023, pp. 127–143.
- 7 S. B. A. Shoilee, “Knowledge discovery for provenance research on colonial heritage objects,” in *Proceedings: Doctoral Consortium at ISWC 2022 co-located with 21st International Semantic Web Conference (ISWC 2022)*, CEUR Workshop Proceedings, 2022.

Journal Articles

- 1 B. Sartini, S. B. Alam Shoilee, C. A. Libbi, and V. De Boer, “Multivocal exhibition: Exploring cultural perspectives through user-curated art exhibitions,” *ACM Journal on Computing and Cultural Heritage*, vol. 17, no. 4, pp. 1–21, 2025.

Book Chapters

- 1 V. de Boer and S. B. A. Shoilee, “Linked open data for cultural heritage,” in *he Palgrave Encyclopedia of Cultural Heritage and Conflict*, I. Saloul and B. Baillie, Eds., Cham: Publisher Name, 2025, pp. 1–7.

Skills and Competencies

Programming Languages	■ Python , C/C++, Java, Node.js Proficient in scripting, data analysis, and backend development.
Web & Data Technologies	■ HTML, CSS, JavaScript, SQL Experienced with front-end prototyping
Semantic Web Technologies	■ SPARQL , SHACL, Data Linking, Data Modeling, KG Constructions Skilled in data retrieval, management, and integration for structured and unstructured data.
AI Techniques	■ Agentic AI, Computer Vision, Information Extraction, Data Mining Theoretical and practical understanding of different approaches.
Tools & Frameworks	■ Git , Docker, LaTeX $\text{\LaTeX}$, HPC cluster Familiar with collaborative tools and semantic annotation environments.
Languages	■ English (fluent), Bengali (native), Dutch (basic)

Awards and Achievements

- 2025 ■ **Best Paper Award**, Second International Workshop of Semantic Digital Humanities (SemDH 2025), co-located with ESWC-2025.
Awarded for the paper “Enhancing Provenance Research with Linked Data: A Visual Approach to Knowledge Discovery.”
- **Nomination for Best Demo**, in Extended Semantic Web Conference (ESWC-2025).
Nominated for the paper “PM-SAMPO: Semantic Portal for Heritage Object Provenance Research”
- 2022 ■ **Polyvocabulary Award**, HackaLOD, Utrecht, the Netherlands.
Received €1500 prize with the Cultural AI Lab team for Linked Data storytelling for museum exhibitions.
- 2018 ■ **SIGCHI Travel Grant**, ACM SIGCHI and HCI Professionals Association of India.
Covered travel, accommodation, and registration for summer school participation.
- 2017 ■ **Merit Excellence Award**, North South University, Dhaka, Bangladesh.
Received 50% tuition waiver for academic excellence in the Master’s program.

Miscellaneous Experience

Research Visits

- Jan-Mar 2025 ■ **Research Visit, Aalto University**, Finland.
Visited SeCo Group, lead by Prof. Eero Hyvönen, for 2.5 months and collaborated on the construction of semantic portals for the knowledge discovery in cultural heritage.
– Implementation of PM-Sampo web application prototype during this visit.
– Two award winning short papers during ESWC-2025 conference.

Hackathons

- November 2022 ■ **Participant, HackaLOD**, Utrecht, the Netherlands.
Participated with a team of PhD researchers from the Cultural AI Lab in a national Linked Data hackathon on cultural heritage.
– Won Polyvocabulary Award for the contribution.
– The work consequentially become a publication at the Journal on Computing and Cultural Heritage (JOCCH).

Miscellaneous Experience (continued)

Summer Schools

- July 2022  **International Semantic Web Research Summer School (ISWS)**, Bertinoro, Italy.
Participated in research training and interdisciplinary group projects on Semantic Web and Linked Data.
- May 2018  **Summer School: Research Methods and Approaches to Text Entry and Other Interaction Techniques**, IIT Bombay, Mumbai, India.
Participated in lectures and hands-on sessions focused on interaction design, usability evaluation, and empirical research methods.

Invited Talks

- February 2024  **Polyvocal Knowledge Modelling for Ethnographic Heritage Object Provenance.**
Cultural AI Event — Open Up Your Collections!
- January 2024  **Guest Lecture, Vrije University Amsterdam**
Course: Digital Humanities and Social Analytics in Practice.
- July 2023  **Talk: Tracing the Unknown: Learning from Provenance Data.**
Conference: Museums, Collections and Society.
- April 2022  **Graph Mining on Historical Past to Digitally Trace Colonial Objects.**
Create Salon at UvA — Polyvocal Perspectives in Archives.

Peer Review Activities

- 2022 – 2025  **Reviewer.**
Conferences: HHAI (2026), ESWC (2026), ISWC (2022 - 2025), EKAW (2024), SemDH Workshop (2024 - 2026).
Journal: Journal on Computing and Cultural Heritage (JOCCH).

Organizational Activities

- 2023 – 2025  **Coordinator and Host, Cultural AI Reading Club**, the Netherlands.
– Organized monthly interdisciplinary reading sessions for researchers.
– Moderated discussions and facilitated engagement with invited scholars across domains.
Occasionally led sessions and discussions in the absence of a host.
- 2021 & 2024  **Volunteered for SEMANTICS Conference Organization**, the Netherlands.

References

Available on Request